

Database Schema

The database schema is structured to store the historical weather forecast data efficiently and is designed with normalization in mind to avoid redundancy. It is secured by IP whitelisting restrictions to VPN server, Scrapy server and API.

Tables

1. City Table

- **Purpose:** Stores information about cities for which weather data is available.
- **Fields:**
 - `id` (Primary Key): Unique identifier for each city.
 - `name`: Name of the city.
 - `country_id`: Foreign Key referencing the Country table.

2. Country Table

- **Purpose:** Stores information about the countries where the cities are located.
- **Fields:**
 - `id` (Primary Key): Unique identifier for each country.
 - `name`: Name of the country.

3. Source Table

- **Purpose:** Stores information about the data source (e.g., MeteoProg, MeteoBlue, etc.).
- **Fields:**
 - `id` (Primary Key): Unique identifier for each source.
 - `name`: Name of the data source.

4. Forecast Table

- **Purpose:** Stores the weather forecast data for a specific city, country, date and source.
- **Fields:**
 - `id` (Primary Key): Unique identifier for each forecast entry
 - `city_id`: Foreign Key referencing the City table
 - `country_id`: Foreign Key referencing the Country table
 - `state`: string with "state name" if necessary
 - `source_id`: Foreign Key referencing the Source table
 - `collection_date`: The date of the collection of the forecast in UTC
 - `forecasted_day`: The date for which the forecast is made in UTC
 - `temp_high`: Day temperature in Celsius

- temp_low: Night temperature in Celsius
- precipitation_chance: Percentage chance of precipitation
- precipitation_amount: Amount of day's expected precipitation in millimeters
- humidity: humidity level for the given day
- wind_speed: in km/h
- weather_condition : description of the weather conditions

Indexes

- ix_forecast_forecasted_day: Speeds up queries filtering by the forecasted date in the forecast table.
- ix_forecast_collection_date: Improves performance for queries filtering by the date when the forecast data was collected.
- ix_forecast_cityid_sourceid_forecasted_day: Optimizes queries filtering by city, source, and forecasted day, useful for retrieving forecasts for specific cities and sources on particular days.
- ix_forecast_cityid_forecasted_day: Enhances query performance for filtering by both city and forecasted day.
- ix_forecast_country_id: Accelerates queries filtering by country, improving performance for country-specific forecast lookups.
- ix_forecast_country_source_id: Speeds up queries filtering by both country and data source, helpful for finding forecasts from specific sources for a given country.
- ix_source_name: Improves performance for queries filtering by the source name, typically used to identify the provider of a specific forecast.